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Comparison of two pasture growth models of differing complexity

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ABSTRACT

Two pasture growth models that shared many common features but differed in model complexity were refined for incorporation into the Integrated Farm System Model (IFSM), a whole-farm model that predicts effects of weather and management on hydrology, soil nutrient dynamics, forage and crop yields, milk or beef production, and farm economics. Major differences between models included the explicit representation of roots in the more complex model and their effects on carbon partitioning and growth. The simple model only simulated aboveground processes. The overall goal was to develop a model capable of representing forage growth and ecosystem carbon fluxes among multiple plant species in pastures while maintaining a relatively simple model structure that minimized the number of required user inputs. Models were compared to observed yield data for 12 site-years from three experiments in central Pennsylvania, USA. Both models underestimated observed yield by 6% when averaged across site-years. However, the simple model provided a better fit to the one-to-one line between observed and simulated yield than did the complex model. The models also showed similar relationships between yield and gross primary productivity (GPP), despite the fact that the complex model was specifically developed to optimize simulation of GPP. The simple model predicted much greater shoot respiration and carbon partitioning to above ground plant tissues, but less shoot senescence than the complex model. Published data on the proportion of GPP consumed in aboveground or total plant respiration exhibit a wide range of values, making it impossible to determine which model provided the best representation of respiration rates and, thus, of the entire carbon budget.

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1. Introduction

The Integrated Farm System Model (IFSM; Rotz et al., 2007) is a deterministic, process-based model that predicts effects of weather and management on hydrology, soil nutrient dynamics, forage and crop yields, milk or beef production, and farm economics in temperate regions at a whole-farm scale. Future enhancements will incorporate whole-farm gaseous emissions, including ammonia and greenhouse gases, into the model framework. Whole-farm models, such as IFSM, provide a means to evaluate the ability of different production systems to optimize plant and animal productivity and economic returns while at the same time delivering ecological goods and services such as increased soil carbon storage and reduced emissions of other greenhouse gasses. To predict whole-farm carbon dynamics, a plant growth model must be able to accurately predict photosynthetic inputs and respiratory losses.

The Integrated Farm System Model (IFSM) was recently enhanced to represent the growth and competition of multiple plant species in pastures and their effects on pasture productivity and botanical composition in temperate climates (Corson et al., 2006). This enhanced model incorporated plant, water, and soil components of the Simulation of Production and Utilization of Rangelands model (SPUR 2.4; Foy et al., 1999). The enhanced model predicted soil water content and biomass yield reasonably well, but did not adequately predict the relative contribution of individual species to total yield (Corson et al., 2006). The model was next modified to improve simulation of carbon inputs (Skinner et al., 2008). Following those revisions, annual GPP could be predicted with a high degree of accuracy, as could overall seasonal patterns in carbon uptake. All these modifications, however, have greatly increased the number of species- and site-specific parameters required to simulate multiple-species pastures.

The level of complexity needed for a specific model depends on the questions being asked and the amount of information available for model building and testing (Boote et al., 1996). Boote et al. (1996) suggested that models should be simplified as much as possible, in part because complex models often require inputs from

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field experiments that are not practical to obtain. On the other hand, complex mechanistic models are essential if the goal is to understand plant processes or formulate hypotheses for experimental testing (Vico and Porporato, 2008). The increase in complexity of the enhanced IFSM made its application to a wide variety of farms unwieldy because of the number of parameters needed to describe each farm. Thus, Corson et al. (2007) developed a simpler, yet reasonably accurate, model of multiple-species pastures for inclusion in IFSM. The revised model required fewer than half the number of physiological parameters to characterize each species, and a greatly reduced number of site-specific parameters compared to the more complex version. Corson et al. (2007) concluded that, given the generality and realism required of IFSM, the degree of precision in the simplified model was acceptable for comparing the effects of different management scenarios on forage productivity.

Other studies have compared the mechanistic representation of physical and biological processes, and predictive ability of plant or soil models of differing complexity. Gao et al. (2004) compared the performance of four photosynthesis models, two of which were highly mechanistic biochemical models and two of which were simplified leaf-level models. They concluded that the more mechanistic biochemical photosynthesis models did not offer significant advantages over the simpler leaf photosynthesis models for describing field data. Likewise, Deen et al. (2003) evaluated four crop-weed competition models (ALMANAC, APSIM, CROPSIM, and INTERCOM) against a common dataset, finding that increased model complexity did not greatly improve model predictions. In contrast, Aber et al. (1996) found that a multiple linear regression model was not able to predict gross carbon exchange by a forest canopy as well as a model based on physiological measurements at the leaf level, suggesting that a certain level of complexity was necessary for modeling carbon assimilation and yield. Similarly, in an exercise that modeled 31 temperate and tropical grasslands, the CENTURY model also performed slightly better than empirical regression equations for predicting plant production and peak live biomass (Hall et al., 1995).

The relative complexity or simplicity of different models is a highly subjective concept. Photosynthesis models can range from simple computations of uptake based on intercepted photosynthetically active radiation (PAR) and canopy radiation use efficiency (Nouvellon et al., 2001), to complex biochemical representations of carbon assimilation processes (Farquhar et al., 1980), which can be further enhanced to included biophysical relationships such as mesophyll conductance and leaf water potential effects on metabolic processes (Vico and Porporato, 2008). Even the simple radiation interception models can become increasing complex as leaf area index, canopy structure, soil albedo, and leaf optical properties are included in simulating intercepted PAR (Nouvellon et al., 2001). Thus, a model considered to be simple in one application could be relatively complex under other circumstances.

The purpose of the current research was to compare the two forage production models developed for IFSM that differed in the complexity of their representations of environmental and management effects on carbon uptake, partitioning among plant compartments, respiratory loss, and forage yield. Most notable among the differences was the lack of explicit representation of roots in the simple model. Designation of these models as "simple" or "complex" was strictly relative to each other, without any attempt to place them within the overall continuum of model complexity. The null hypothesis was that no difference existed between the two models in their ability to simulate key components of the plant carbon cycle. Accurate simulation of plant carbon dynamics is a necessary step in providing the simulations of pasture and whole-farm carbon fluxes that are needed by producers and policy makers to guide greenhouse gas reduction efforts.

2. Methods

2.1. Model development

Development of the two pasture models has been described in detail elsewhere (Corson et al., 2006, 2007; Skinner et al., 2008). Briefly, the IFSM has been modified to simulate multiple functional groups with the pasture system. In these simulations, functional groups are based on morphological characteristics and include grasses, legumes, and non-leguminous forbs. The complex model was based on the rangeland model, SPUR, which was modified to represent temperate pasture conditions. In the complex model, root and shoot nitrogen and carbon structural and substrate contents, of both live and dead tissue, were represented explicitly, as were the nitrogen and carbon contents of plant residue, soil-applied manure, and soil organic matter. Plants recycled a portion of structural carbon from dying shoot and root tissue to substrate-carbon pools. Transpiration from each soil layer was a function of plant rooting depth and was taken first from the upper layer, with unmet demand cascading to the next lower layer.

The complex model was originally calibrated with data from the three-species mixtures from the 2002 Dairy Graze experiment (Table 1) (Corson et al., 2006), then the photosynthesis component was refined based on data from the Haller Grass 2003 experiment that contained a single functional group (Skinner et al., 2008). To refine the photosynthesis model, daytime fluxes were measured by eddy covariance then partitioned into their photosynthetic and respiration components as described in Gilmanov et al. (2003). Continuous micrometeorological measurements of photosynthetic uptake at the Haller Grass site allowed seasonal and even daily comparisons between observed and simulated plant carbon assimilation.

In contrast, the simplified pasture growth model, based on the Grazing Simulation Model (GRASIM) (Mohtar et al., 1997), lacked an explicit representation of roots. A portion of daily assimilated carbon was partitioned belowground by the model but its fate was not tracked beyond that point. Soil organic matter was represented only as biomass, from which ammonium was mineralized. Dying plant tissue did not recycle structural carbon to substrate-carbon pools. Soil was divided into four layers. Transpiration from each of the first three soil layers was fixed, and all unmet transpiration demand came from the lowest layer. Compared to the complex model, the simple model required considerably fewer species- and site-specific parameters and variables (Table 2). This model was calibrated with data from three-species mixtures from the 2002 Dairy Graze experiment.

Table 1
Brief description of study sites used for comparison of pasture growth models

Site	Year	Functional groups (no.)	Harvests (no.)	Precipitation (mm)	Forage Yield (g m^{-2})
Haller Grass	2003	1	4	1108	403
Haller Grass	2004	1	4	1190	358
Haller Grass	2005	1	4	715	205
Haller Alfalfa	2003	2	3	1108	478
Haller Alfalfa	2004	2	3	1190	319
Haller Alfalfa	2005	1	3	715	209
GRACEnet simple	2005	2	5	715	374
GRACEnet complex	2005	3	5	715	432
Dairy Graze 2 species	2002	2	5	942	432
Dairy Graze-2 species	2003	2	7	1148	804
Dairy Graze-3 species	2002	3	5	942	538
Dairy Graze-3 species	2003	3	7	1148	819

Pastures contained either 1 (grass), 2 (grass–legume), or 3 (grass–legume–forb) functional groups based on plant morphological traits.

Table 2
Parameters and initial variables required by the complex version of the pasture submodel of the Integrated Farm System Model (IFSM)

Species-specific parameter/variable	Site-specific parameter/variable
Photosynthetic rate (maximum or mean)	**Condition I curve number for site**
Light-use efficiency coefficient	**Soil evaporation factor**
Maximum temperature for photosynthesis	**Number of soil layers**
Optimum temperature for photosynthesis	**Moist bulk density of soil layers**
Minimum temperature for photosynthesis	**Porosity of soil layers**
Soil-water influence on photosynthetic rate	**Soil water content at drained upper limit**
Specific leaf area	**Soil water content at lower limit**
Maximum nitrogen uptake coefficient	**Saturated soil-hydraulic conductivity/ drainage rate**
Canopy-level light extinction coefficient	**Bare soil albedo**
Maximum symbiotic N fixation rate	**Thickness of soil layers 1–5**
Temperature for frost kill	**Percent silt in soil**
Optimal temperature for senescence	**Percent clay in soil**
Maximum rooting depth	**Initial inorganic nitrate in upper soil layer**
Initial live shoot dry matter	**Initial inorganic nitrate in lower soil layer**
Maximum nitrogen concentration in live shoots	**Initial inorganic ammonium in upper soil layer**
Shoot maximum specific growth rate	**Initial plant litter dry matter**
	Rate of nitrate denitrification
Proportion of phytomass susceptible to trampling	**Mineralization rate of organic manure residue nitrogen**
Stocking rate tolerance of standing dead	**Mineralization rate of organic plant residue nitrogen**
Stocking rate tolerance of green shoots	**Soil leaching coefficient**
Proportion of green shoots susceptible to death	**Area of pasture**
Maximum root:shoot ratio	**Number of grazing cows**
Proportion of root phytomass translocated to shoots	**Proportion soil N unavailable to plants**
Germination proportion	
Proportion additional shoot death after senescence	Watershed parameter representing slope and size relationships
Seed and root mortality proportions	Watershed parameter representing climatic characteristics
Root and shoot respiration proportion	Watershed parameter representing hydrograph shape
Nitrogen use efficiency coefficient	Proportion of bare soil surface
Maximum proportion of shoot structure that photosynthesizes	Initial soil water content as proportion of drained upper limit
Maximum leaf area index	Dead root dry matter in soil
Temperature for root translocation to shoot	Decomposition rate of dead root dry matter
Minimum water potential for root translocation to shoot	Decomposition rate of soil organic matter
Minimum water potential for seed germination	Soil erosion calculated with MUSLE[a]
Day of year seed production begins	
Days of year senescence begins and ends	
Initial live root dry matter	
Initial propagule dry matter	
Initial dead shoot dry matter	
Root distribution exponent	
Root maximum specific growth rate	
Precipitation tolerance coefficient	
Proportion of photosynthate translocated to roots after senescence begins	
Proportion of photosynthate translocated to propagules after flower initiation	

The reduced set of parameters required by the simple version is shown in boldface.
[a] Modified Universal Soil Loss Equation.

The IFSM was designed for repeated simulations of a single year against a range of environmental conditions. Thus, results from one year have no impact on simulations for subsequent years. The proportion of aboveground plant material in each functional group (i.e. grass, legume, or forb) was determined from observed data in May of each year. Plants were then allowed to compete with each other for light, water, and nutrients so that the proportional contribution of each functional group to total biomass could change within each year. Harvests, whether for hay or grazed, were simulated on the actual date they occurred, and the biomass remaining following harvests was obtained from observed data to determine the starting point for the next growth cycle. Nitrogen fertilizer inputs were determined from farm records, whereas, manure deposition was calculated based on the amount of consumed forage as simulated by the models. Soil samples from the beginning of each experiment were used to input soil organic matter content.

2.2. Grazing experiments

The Haller Grass and Haller Alfalfa sites (Table 1) were located on two pastures at the Pennsylvania State University Haller Research Farm located about 10 km northeast of State College, Pennsylvania (40.9° N; 77.8° W). Soil type was a Hublersburg silt loam (Typic Hapludult; FAO classification, Haplic Acrisols) with 3–8% slopes. Both locations had been sown to perennial forage species since 1968. The grass-based pasture was last reseeded in 1982 and was dominated by a mixture of cool-season grasses including orchardgrass (*Dactylis glomerata* L.), tall fescue (*Festuca arundinacea* Schreb.), and Kentucky bluegrass (*Poa pratensis* L.). The pasture contained other common species including smooth bromegrass (*Bromus inermis* Leyss.), dandelion (*Taraxacum officinale* L.), and alfalfa (*medicago sativa* L.). The Haller Grass pasture was simulated as a one-functional group (grass) pasture for all three years.

The alfalfa-based pasture was planted as an alfalfa monoculture in 1995. Intermixed with the alfalfa in 2003 were patches of orchardgrass, smooth bromegrass, dandelion, Kentucky bluegrass, and tall fescue. The proportion of alfalfa decreased from about 50% to 75% of plant cover in 2003 to <5% in 2005, possibly because of wet conditions in 2003 and 2004. By 2005, the predominant species in the alfalfa pasture were reed canarygrass (*Phalaris arundinacea* L.) and orchardgrass. Therefore, the Haller Alfalfa site was simulated as a two-functional group, grass/legume pasture in 2003 and 2004, and as a one-functional group, grass pasture in 2005.

Pastures were harvested three to four times per year between mid-May and mid-November. The grass-based pasture was cut once for hay in May 2003, with all subsequent biomass removal by grazing. In this and all other experiments, management intensive rotational grazing was employed wherein paddocks were grazed for 1–3 days depending on size of the paddock and number of cattle available. Amount of harvested biomass was determined by measuring available forage immediately before and after harvests. The alfalfa-based pasture was cut for hay once or twice in the spring and early-summer each year then grazed one to three times in the late-summer and autumn. The grass-based pasture received N fertilizer as urea twice each year at rates of 56 kg N ha^{-1} in April and 34 or 45 kg N ha^{-1} in August. The alfalfa-based pasture was not fertilized.

The GRACEnet pastures (Table 1) were part of a nation-wide USDA-ARS effort to identify management systems that minimize net greenhouse gas emissions (Jawson et al., 2005). Pastures were located immediately adjacent to the Haller Grass site, and thus, shared the same soil characteristics and weather conditions. The pre-existing alfalfa crop at the site was killed in October 2003, and rye (*Secale cereale* L.) (November–March) followed by oat (*Avena sativa* L.) (April–July) cover crops were planted prior to pasture establishment in August 2004. Plots received 3.36 Mg/ha high magnesium lime, 10 kg/ha N, 54 kg/ha P, and 236 kg/ha K in March 2004. The simple pasture mixture consisted of a two-species/two-functional group (grass–legume) mixture of orchardgrass and white clover (*Trifolium repens* L.). The complex mixture consisted of five species/three-functional groups (grass–legume–forb) including orchardgrass, tall fescue, alfalfa, white clover, and chicory (*Cichorium intybus* L.). Pastures were grazed five times in 2005 between late-May and late-October. No additional fertilizer was added to the plots following the initial application in March 2004.

The Dairy Graze plots were located at the Pennsylvania State University, Dairy Cattle Research and Education Center located about 9 km southwest of the Haller Research Farm. Soil at the site is a Hagerstown silt loam (fine, mixed, semi-active, mesic, Typic Hapludalfs; FAO classification, Chromic Luvisols). In July 2001 existing vegetation at the site was killed and four forage mixtures containing two, three, six, or nine species were no-till seeded in replicated 1-ha pastures on 29 August 2001. Soil tests in 2001 indicated that pH, available P, and available K were all adequate for forage production; thus, no fertilizer was applied. The pastures were subdivided into smaller paddocks and rotationally grazed five to seven times between April and September of 2002 and 2003 (Sanderson et al., 2005). Simulations were run for the two-species/functional groups (orchardgrass–white clover) and three-species/functional groups (orchardgrass–white clover–chicory) mixtures.

2.3. Model evaluation

The square root of the mean squared deviation (SQRT MSD) from the 1:1 line between predictions and observations, and its subcomponents (Gauch et al., 2003), was used to evaluate model success in predicting annual yield and seasonal yield distribution. Models are most successful when SQRT MSD is low and the greatest proportion of the variability is due to scatter about the 1:1 line rather than to systematic over- or under-prediction (translation) or to deviations in slope from unity (rotation).

Observed data on individual flux components, including photosynthesis, respiration, and root/shoot partitioning were not available with the exception of photosynthesis data for the Haller Grass 2003, 2004, 2005, and Haller Alfalfa 2003 sites. Because the photosynthetic component of the complex model was calibrated and validated against the Haller data, and the simple model was not, comparison of the models against these data would introduce a bias in favor of the complex model. Therefore, comparisons between models for these parameters were made relative to each other, without reference to observed data.

3. Results

Both models provided robust predictions of annual forage yield (Fig. 1). Mean (±1 SE) absolute differences between simulated and observed annual yields were 19 ± 4% and 28 ± 10% for the complex and simple models, respectively. With a few exceptions, the relationship between simulated and observed yield for both models fell close to the 1:1 line for all years and sites. The SQRT MSD for the simple model was 127 g m^{-2} compared with 163 g m^{-2} for the complex model. Ninety-five percent of the variability in SQRT MSD for the simple model was due to scatter about the 1:1 line. For the complex model, rotation about the axis made a significant contribution (38%) to total variability in SQRT MSD, with a corresponding reduction in the contribution of scatter about the 1:1 line (60%). Neither model showed a tendency to systematically over- or under-estimate yield.

The simple model overestimated yield by 133% for the Haller Grass site in 2005 (Tables 3 and 4), which was the driest year of the study (Table 1). Eliminating that one site would reduce the difference between simulated and observed yield for the simple model from 28 ± 10% to 18 ± 3%. The complex model provided poor estimates of yield for both the Dairy Graze sites in 2003, which had the greatest yields of any site or year, underestimating yield by 51% for the two-species mixture while overestimating yield by 41% for the three-species mixture. If those sites were eliminated, the difference between simulated and observed yield for the complex model would be reduced to 14 ± 3%, and the contribution of rotation about the 1:1 line would be eliminated, resulting in nearly identical slopes and intercepts for the complex and simple models.

No difference between models existed in the relationship between simulated GPP and simulated yield (Fig. 2), and 84% of the variability in annual yield could be explained by differences in annual CO_2 assimilation. However, the simple model partitioned a greater proportion of GPP to aboveground structural dry matter and shoot respiration than did the complex model (Fig. 3). Averaged across simulations, 33% of GPP was partitioned aboveground in the simple model compared with 28% for the complex model. Substantial differences also existed between the models in the relationship between aboveground respiration and GPP (Fig. 4).

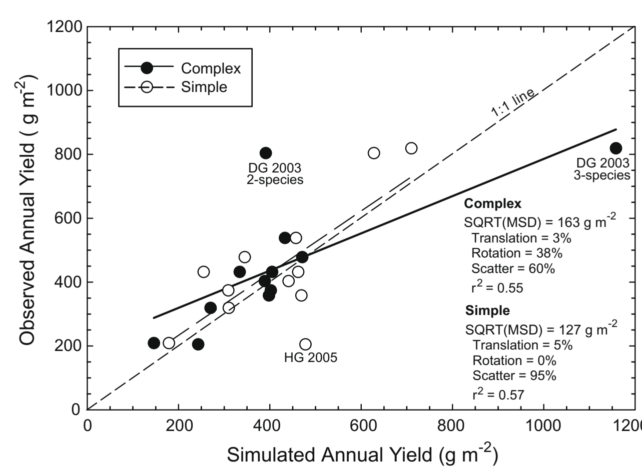

Fig. 1. Observed annual forage yield vs. yield predicted by the simple and complex pasture models. Data points with the greatest percentage difference between observed and simulated values are identified on the graph for Haller Grass (HG) 2005, and Dairy Graze (DG) 2003. Goodness-of-fit was evaluated based on the square root of the mean squared deviation (SQRT MSD) from the 1:1 line between predictions and observations, and its subcomponents:deviation about the 1:1 line (scatter); systematic over- or under-prediction (translation); and deviation in slope from unity (rotation).

Table 3
Observed vs. simulated yield, and goodness-of-fit for the complex and simple models based on the number of plant functional groups simulated

Site	# groups[a]	Complex Model					Simple Model				
		Yield	SQRT MSD	Translation	Rotation	Scatter	Yield	SQRT MSD	Translation	Rotation	Scatter
		%[b]	$g\ m^{-2}$	% of MSD			%	$g\ m^{-2}$	% of MSD		
Haller Grass 2003	1	3	15	5	16	80	9	42	5	4	91
Haller Grass 2004	1	12	50	4	0	96	31	84	11	40	49
Haller Grass 2005	1	19	19	26	1	73	133	93	54	44	2
Haller Alfalfa 2005	1	30	83	6	91	3	14	91	1	66	33
Mean		**16**	**42**	**10**	**27**	**63**	**45**	**78**	**18**	**39**	**44**
Haller Alfalfa 2003	2	1	90	0	2	98	28	76	34	26	39
Haller Alfalfa 2004	2	15	18	78	14	8	3	112	0	97	3
GRACEnet-Simple	2	7	35	3	60	37	17	23	34	12	55
Dairy Graze-2 2002	2	6	86	0	21	79	7	91	0	29	71
Dairy Graze-2 2003	2	51	80	41	1	58	22	74	9	26	65
Mean		**16**	**62**	**24**	**20**	**56**	**15**	**75**	**15**	**38**	**47**
GRACEnet-Complex	3	23	38	7	54	39	34	35	55	0	45
Dairy Graze-3 2002	3	20	91	4	47	49	15	81	3	36	61
Dairy Graze-3 2003	3	41	131	10	74	16	134	72	4	375	595
Mean		**28**	**87**	**7**	**58**	**35**	**21**	**63**	**21**	**24**	**55**

Goodness-of-fit was evaluated based on the square root of the mean squared deviation (SQRT MSD) from the 1:1 line between predictions and observations, and its subcomponents: deviation about the 1:1 line (scatter); systematic over- or under-prediction (translation); and deviation in slope from unity (rotation).
[a] Number of plant functional groups simulated.
[b] Absolute value of percentage difference between observed and simulated annual yield.

Table 4
Observed vs. simulated yield and goodness-of-fit for the complex and simple models based on annual number of harvests

Site	# of Harvests	Complex Model					Simple Model				
		Yield	SQRT MSD	Translation	Rotation	Scatter	Yield	SQRT MSD	Translation	Rotation	Scatter
		%[a]	$g\ m^{-2}$	% of MSD			%	$g\ m^{-2}$	% of MSD		
Haller Alfalfa 2003	3	1	90	0	2	98	28	76	34	26	39
Haller Alfalfa 2004	3	15	18	78	14	8	3	112	0	97	3
Haller Alfalfa 2005	3	30	83	6	91	3	14	91	1	66	33
Mean		**15**	**64**	**28**	**36**	**36**	**15**	**93**	**12**	**63**	**25**
Haller Grass 2003	4	3	15	5	16	80	9	42	5	4	91
Haller Grass 2004	4	12	50	4	0	96	31	84	11	40	49
Haller Grass 2005	4	19	19	26	1	73	133	93	54	44	2
Mean		**11**	**28**	**12**	**6**	**83**	**58**	**73**	**23**	**29**	**47**
GRACEnet-S 2005	5	7	35	3	60	37	17	23	34	12	55
GRACEnet-C 2005	5	23	38	7	54	39	34	35	55	0	45
Dairy Graze-2 2002	5	6	86	0	21	79	7	91	0	29	71
Dairy Graze-3 2002	5	20	91	4	47	49	15	81	3	36	61
Mean		**14**	**63**	**4**	**46**	**51**	**18**	**58**	**23**	**19**	**58**
Dairy Graze-2 2003	7	51	80	41	1	58	22	74	9	26	65
Dairy Graze-3 2003	7	41	131	10	74	16	13	72	4	37	59
Mean		**46**	**106**	**26**	**38**	**37**	**18**	**73**	**7**	**32**	**62**

Goodness-of-fit was evaluated based on the square root of the mean squared deviation (SQRT MSD) from the 1:1 line between predictions and observations, and its subcomponents: deviation about the 1:1 line (scatter); systematic over- or under-prediction (translation); and deviation in slope from unity (rotation).
[a] Absolute value of percentage difference between observed and simulated annual yield.

For a given level of yield, shoot respiration was always greater in the simple compared with the complex model. On average, 18% of GPP was consumed by shoot respiration in the simple model compared with 10% in the complex model.

When respiration was partitioned into growth and maintenance components, average growth respiration rates of 255 and 330 g CO_2 m^{-2} yr^{-1} were predicted for the simple and complex models, respectively. However, maintenance respiration was much greater in the simple model, averaging 739 g CO_2 m^{-2} yr^{-1} compared with 232 g CO_2 m^{-2} yr^{-1} for the complex model. Even though respiration rates differed, both models showed highly significant relationships between GPP and total shoot respiration ($r^2 = 0.91$, $P < 0.01$ for each model). Shoot senescence rates were much higher in the complex model, averaging 387 g dry matter m^{-2} yr^{-1} vs. 97 g dry matter m^{-2} yr^{-1} for the simple model. Aboveground litter production in the complex model was nearly as great as the harvested biomass which averaged 420 g dry matter m^{-2} yr^{-1}. Differences in growth respiration between models were due to differences in total aboveground biomass production, i.e. harvested biomass plus litter.

Models tended to be most successful at simulating within-year seasonal yield distribution for pastures that contained a similar number of functional groups as the pastures that were used to calibrate the model (Table 3). For each model, SQRT MSD increased and the percentage of MSD due to scatter decreased as the number of functional groups increasingly deviated from the number of functional groups for which the model was calibrated. The complex model had smaller SQRT MSD (42 vs. 63 g m^{-2}) and greater scatter (63% vs. 55%) than the simple model when conditions similar to their respective calibration data sets were compared, i.e. a single functional group for the complex model vs. three-functional groups for the simple model. However, when the complex model

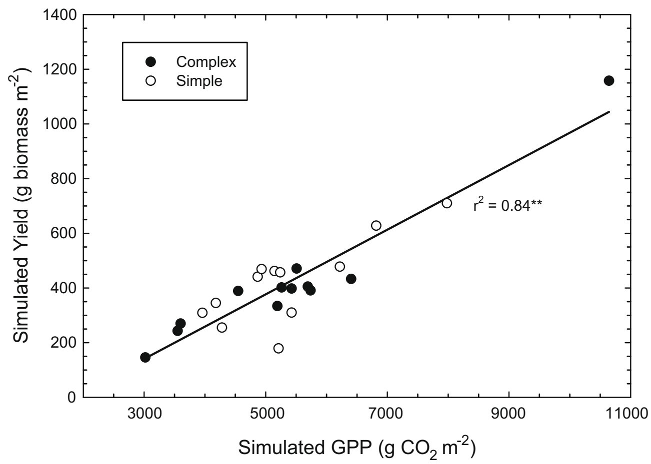

Fig. 2. Simulated annual gross primary productivity (GPP) vs. simulated annual forage yield for simple and complex pasture models. ** Indicates significance at $P < 0.01$.

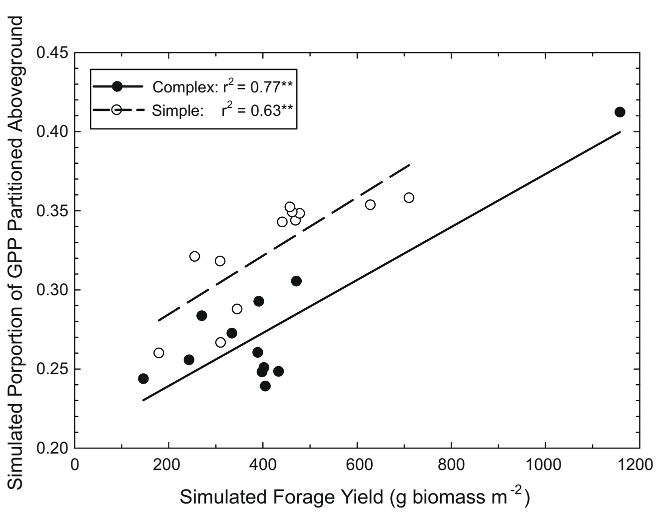

Fig. 3. Simulated portion of annual gross primary productivity (GPP) partitioned aboveground vs. simulated annual forage yield in the simple and complex pasture models. ** Indicates significance at $P < 0.01$.

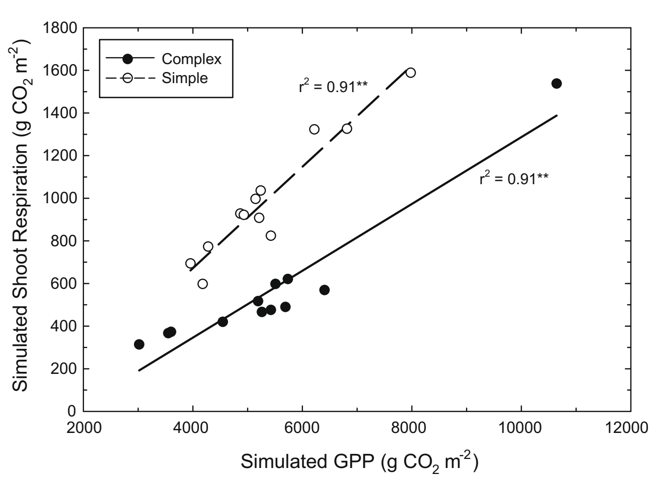

Fig. 4. Simulated annual gross primary productivity (GPP) vs. simulated shoot respiration in the simple and complex pasture models. ** Indicates significance at $P < 0.01$.

was used to simulate pastures with three-functional groups, SQRT MSD was greater (87 vs. 78 g m^{-2}) and scatter less (35% vs. 44% of MSD) than when the simple model was used to simulated one-functional group.

The number of times a pasture was harvested each year also affected the ability of both models to accurately simulate within-year yield distribution. The complex model was calibrated for a year when four harvests occurred compared with five harvests during the calibration year for the simple model. Predictions were most accurate when the number of harvests simulated was similar to the number of harvests for which the model was developed (Table 4). The complex model again performed better than the simple model when conditions were similar to their calibration data sets (SQRT MSD = 28 vs. 58 g m^{-2}, scatter = 83% vs. 58% of MSD). However, performance of the complex model became worse than the simple model as the number of harvests deviated from the number of harvests during the calibration year.

4. Discussion

Both the simple and complex pasture models provided reasonable predictions of yield and GPP. Although overall performance was good, important insights can be gained by examining where the models performed poorly. The complex model had the greatest difficulty simulating yield from the Dairy Graze study in 2003. In both the two- and three-functional group pastures, the model greatly underestimated observed yield during the first two growth periods. Observed yields in the Dairy Graze study were much greater during early-spring than in other experiments and both models had difficulty predicting growth rates as great as those observed. Weiss and Wilhelm (2006) have pointed out that researchers must also be concerned about the accuracy of experimental data and environmental inputs when comparing simulated and observed values. For several years prior to the Dairy Graze study the site had been irrigated with nutrient rich water from the Penn State waste treatment facility. It was difficult to properly quantify these inputs and their effects on soil fertility, or determine how best to include them in the site-specific parameters within the models. Thus, underestimation of observed yields could have resulted from an inability to adequately describe initial pasture conditions under this unique situation rather than from structural problems within the models.

Despite the underestimation of forage yield during the first two growth cycles in the three-functional group Dairy Graze pasture in 2003, the complex model overestimated total annual yield by 40%. All of the overestimation resulted from the simulated legume component and occurred during August and September, but especially in September, when simulated yield exceeded observed yield by more than 300%. Three primary causes for the overestimation were identified. First, residual biomass following defoliation was higher in this particular treatment than in any other treatment or experiment. The resulting large leaf area index led to a more rapid regeneration of the simulated canopy following defoliation than was observed in the field. Second, the final harvest occurred in early-August, which was earlier than other experiments that were typically grazed into September and October. This early defoliation, combined with the high initial leaf area index, allowed for an unusually long period of uninterrupted rapid biomass accumulation during the fall. Finally, autumn growth in simulated grass-dominated pastures was typically constrained by severe nitrogen limitations. No such limitation existed for the legume component and there were no other significant stresses in the autumn of 2003 to limit growth. An additional contributing factor could be the fact that as yield and GPP increased, partitioning aboveground also increased (Fig. 3). The greater aboveground partitioning could have further stimulated aboveground growth in a positive feedback loop.

Although development of the complex model included concerted efforts to optimize predictions of gross photosynthesis as well as yield, and the model had been shown to predict GPP to within 7% of observed values (Skinner et al., 2008), both models ultimately showed similar relationships between GPP and yield (Fig. 2). Thus, increasing model complexity did not improve predictions of GPP. Model complexity can be reflective of a variety of different model characteristics. The relative simplicity or complexity of a model can result from the degree of mechanistic representation of the processes being simulated, from the number of processes being simulated, or from the number and availability of inputs needed to execute the model. For example, Thornley and Cannell (1997) suggested that the Hurley Pasture Model was complex because it included many processes, none of which was regarded as redundant. It is possible that an empirical multiple regression model of a particular process could conceivably require more, and harder to obtain, inputs and thus be more "complex" in certain ways than a highly mechanistic functional model.

In the current study, representations of the processes controlling photosynthesis and shoot respiration were equally mechanistic in the simple and complex models. In fact, the equations for determining growth respiration were identical. Simulations of maintenance respiration also used similar concepts and equations but with different quantitative responses to temperature. Much of the difference between models related to what was not simulated by the simple model, i.e. the cycling of carbon and nitrogen between above and belowground tissues, as well as carbon loss from the root system through senescence and respiration (Table 2).

Functional equilibrium theory suggests that root:shoot partitioning depends on maintaining a balance between root growth and water and nutrient uptake on one hand, and shoot growth and carbon assimilation on the other (Davidson, 1969). Therefore, having a correct representation of root processes might be thought of as essential for modeling carbon partitioning and shoot growth and numerous papers have been written on modeling partitioning between roots and shoots. However, even though basic mechanistic representations of root growth and nutrient uptake might seem scientifically desirable, van Noordwijk and van de Geijn (1996) suggested that for practical purposes simpler models might be satisfactory or even preferable from the point of view of robustness and ease of parameter estimation.

During development of the complex model, changes in maximum root:shoot ratio, root respiration and turnover rates, and in rules governing translocation of nonstructural carbohydrates to roots all affected GPP and yield. However, changing partitioning to roots did not always have the expected effect on aboveground processes. For example, reducing maximum root:shoot ratio from 5:1 to 1:1 gave the expected result of decreasing root structural biomass by 56% while increasing shoot structural biomass by 57%. However, GPP also decreased by 29% despite the increase in aboveground biomass. The lower GPP resulted from increased feedback inhibition of photosynthesis due to greater nonstructural carbohydrate concentration in leaves, and from increased N limitation due to dilution of leaf N as a result of the increased shoot growth.

The inclusion of roots in the complex model increased the model's ability relative to the simple model to correctly simulate yield and GPP when conditions were similar to those for which the model was calibrated, i.e. for a similar number of plant functional groups and harvests (Table 3 and 4). However, the simple model appeared to provide more stable predictions over a range of environmental and management conditions. Passioura (1996) suggested that mechanistic models are often flawed by being based on untestable guesses about the processes being simulated. Proper quantification of those processes can also be problematic. Limited knowledge of root growth, respiration, and turnover rates in the field certainly limited the ability of the complex model to correctly simulate those processes over a wide range of pasture conditions. The relative success of the simple model suggests that model developers should guard against over-parameterizing models with detail unnecessary for predicting variables of interest. Doing so may result in diminishing returns for model users.

The most significant difference between these models was in their predictions of the proportion of GPP consumed by aboveground respiration, and in simulations of aboveground senescence rates. Despite the rather large differences, it was difficult to determine which model provided results that most correctly simulated processes occurring in the field. Cannell and Thornley (2000) suggested, as a rule-of-thumb, that growth and maintenance respiration are approximately equal when averaged over a season. In the current simulations, maintenance respiration exceeded growth respiration by threefold for the simple model, whereas, maintenance respiration in the complex model was only two-thirds of growth respiration. Thus, maintenance respiration probably needed to be increased somewhat in the complex model, but greatly decreased in the simple model. Such differences between models for maintenance respiration are not unusual and the carbon costs of maintenance respiration among published models are highly variable, mostly due to the wide range of possible rate coefficients found in the literature (Thornley and Cannell, 2000).

Respiration is often poorly represented in plant growth models compared with photosynthesis (Cannell and Thornley, 2000). Farrar (1985) suggested that plants respire up to 70% of the carbon accumulated during their lifetime. However, based on theoretical considerations of the costs of growth and maintenance respiration, Amthor (2000) suggested that the allowable range could be anywhere from 35% to 80%. Experimental results for crop and grassland species cited by Amthor (2000) ranged from 34% to 65%. Short-term (weekly, monthly, or seasonal) tracer studies often find about 30–55% of fixed carbon is respired (McCree and Troughton, 1966; Atkinson and Farrar, 1983; Belanger et al., 1994; Saggar et al., 1997). Annual respiration of perennial species will typically be greater than growing-season respiration because of continued metabolic requirements during winter months when photosynthesis is low or non-existent.

Whole-plant respiration was simulated in the complex but not the simple model because the latter lacked an explicit representation of roots. In the complex model, 47% of GPP was respired, which was well within the allowable range suggested by Amthor (2000) and observed by others for grassland species. Given the greater shoot respiration rates in the simple compared to the complex model, it is likely that whole-plant respiration would have also been greater for the simple model if roots had also been included, but would likely still have been within the allowable range.

In a controlled chamber study of two montane grasses, Atkinson and Farrar (1983) found that 41–56% of fixed C was partitioned aboveground, with 23–50% of that C used for respiration and the rest for shoot growth. In field studies, partitioning to shoots comprised 50–60% of assimilated carbon in New Zealand pastures (Stewart and Metherell, 1999), 65–75% for tall fescue (*F. arundinacea* Schreb.) growing in France (Belanger et al., 1994) and 40% for short-grass steppe species of northern Colorado (Milchunas and Lauenroth, 1992). Water and nutrient stress typically increased carbon partitioning to roots in the cited studies.

Mean aboveground partitioning in the complex model was 28% of GPP compared with 33% for the simple model. Both were lower than reported values from the literature cited above for perennial species. This was despite efforts in the development and parameterization of the complex model to reduce belowground partitioning. Aboveground partitioning was closer to that observed in the short-grass steppe (Milchunas and Lauenroth, 1992), which would be expected for the complex model since its plant growth components originated in a rangeland model (Corson et al., 2006). Growth and photosynthesis equations in the simple model originally came from the Grazing Simulation Model (GRASIM, Mohtar et al., 1997) which was specifically developed for temperate pastures. Thus, it is not surprising that greater aboveground partitioning was observed in the simple model since temperate grasslands typically have lower root:shoot ratios than rangelands. However, it appears both models require additional adjustment to increase partitioning to aboveground tissues. Increasing aboveground partitioning in either model would also require increasing shoot respiration and/or senescence rates to retain accurate yield predictions.

Litter (senesced aboveground material not included in yield estimates) comprised 48% of total aboveground biomass in the complex model compared with 19% in the simple model. In a Swiss grassland, litter accounted for 40–80% of orchardgrass aboveground biomass depending on productivity of the site and time of year (Schlapfer and Ryser, 1996). In a Canadian mixed-grass prairie, litter accounted for nearly 50% of aboveground biomass (Willms et al., 1993). However, unpublished data from the locations used in the current study found that dead plant matter accounted for 14–32% of aboveground biomass (Skinner, unpublished data; Sanderson, personal communication). Thus, the complex model appeared to be more representative of published literature values, whereas, the simple model most correctly represented the observed data. Correctly predicting litter accumulation is crucial for simulating ecosystem carbon budgets because of the contribution of aboveground litter to soil organic matter.

Additional field studies that partition ecosystem respiration into its soil microbial, shoot, and root components are urgently needed to provide the data sets necessary to evaluate simulated plant respiratory losses. Unfortunately, separating root from soil microbial respiration is not a trivial undertaking and root respiration has been estimated to account for anywhere from 10% to 90% of total soil respiration depending on methods used, vegetation type and time of year (Hanson et al., 2000). Root growth and turnover are equally poorly understood (Gill et al., 2002), further complicating our ability to simulate root systems as part of whole-plant growth models. Our limited quantitative understanding of root processes and their interactions with aboveground growth probably explains why a simulation model lacking roots was equally capable of simulating photosynthesis, shoot growth, and respiration as a model that included roots.

5. Conclusions

A simple pasture growth model lacking explicit representation of roots was compared with a more complex model that simulated processes associated with carbon and nitrogen transfer between shoots and roots, root respiration, and root turnover. No difference existed between models in their ability to simulate photosynthetic carbon uptake and forage yield. However, significant differences existed in their representation of aboveground partitioning, shoot respiration rates, and litter production. Differences in predictions were not due to inherent mechanistic differences but were rather due to quantitative relationships among model parameters, suggesting that the models could be easily brought into agreement with each other. Currently, large uncertainty among validation datasets from field experiments and in the literature prevents any satisfactory determination of which model most correctly represented the processes controlling respiration and senescence. When predictions of photosynthesis and yield were the primary concern, the simple model was preferred because of its reduced requirement for input data. The simple model was also of appropriate complexity for integration with the other components of the IFSM. However, until better validation data for the processes controlling carbon loss can be collected, it is not clear which model will provide the best representation of total pasture carbon fluxes and will be most appropriate for inclusion in models that simulate the whole-farm carbon budget.

References

Aber, J.D., Reich, P.B., Goulden, M.L., 1996. Extrapolating leaf CO_2 exchange to the canopy: a generalization model of forest photosynthesis compared with measurements by eddy correlation. Oecologia 106, 257–265.

Amthor, J.S., 2000. The McCree–de Wit–Penning de Vries–Thornley respiration paradigms: 30 years later. Ann. Bot. 86, 1–20.

Atkinson, C.J., Farrar, J.F., 1983. Allocation of photosynthetically-fixed carbon in *Festuca ovina* L. and *Nardus stricta* L.. New Phytol. 95, 519–531.

Belanger, G., Gastal, F., Warembourg, F.R., 1994. Carbon balance of tall fescue (*Festuca arundinacea* Schreb.): effects of nitrogen fertilization and the growing season. Ann. Bot. 74, 653–659.

Boote, K.J., Jones, J.W., Pickering, N.B., 1996. Potential uses and limitations of crop models. Agron. J. 88, 704–716.

Cannell, M.G.R., Thornley, J.H.M., 2000. Modeling the components of plant respiration: some guiding principles. Ann. Bot. 85, 45–54.

Corson, M.S., Rotz, C.A., Skinner, R.H., Sanderson, M.A., 2007. Adaptation and evaluation of the Integrated Farm System Model to simulate temperate multiple-species pastures. Agric. Sys. 94, 502–508.

Corson, M.S., Skinner, R.H., Rotz, C.A., 2006. Modification of the SPUR rangeland model to simulate species composition and pasture productivity in humid temperate regions. Agric. Sys. 87, 169–191.

Davidson, R.L., 1969. Effect of root/leaf temperature differentials on root/shoot ratios in some pasture grasses and clover. Ann. Bot. 33, 561–569.

Deen, W., Cousens, R., Warringa, J., Bastiaans, L., Carberry, P.S., Rebel, K., Riha, S., Murphy, C., Benjamin, L.R., Cloughley, C., Cussans, J., Forcella, F., Hunt, T., Jamieson, P., Lindquist, J., Wang, E., 2003. An evaluation of four crop:weed competition models using a common data set. Weed Res. 43, 116–129.

Farquhar, G.D., Von Caemmerer, S., Berry, J.A., 1980. A biochemical model of photosynthetic CO_2 assimilation in leaves of C_3 species. Planta 149, 78–90.

Farrar, J.F., 1985. The respiratory source of CO_2. Plant Cell Environ. 8, 427–438.

Foy, J.K., Teague, W.R., Hanson, J.D., 1999. Evaluation of the upgraded SPUR model (SPUR 2.4). Ecol. Model 118, 149–165.

Gao, Q., Zhang, X., Huang, Y., Xu, H., 2004. A comparative analysis of four models of photosynthesis for 11 plant species in the Loess Plateau. Agric. Forest Meteorol. 126, 203–222.

Gauch, H.G., Hwang, J.T.G., Fick, G.W., 2003. Model evaluation by comparison of model-based predictions and measured values. Agron. J. 95, 1442–1446.

Gill, R.A., Kelly, R.H., Parton, W.J., Day, K.A., Jackson, R.B., Morgan, J.A., Scurlock, J.M.O., Tieszen, L.L., Castle, J.V., Ojima, D.S., Zhang, X.S., 2002. Using simple environmental variables to estimate belowground productivity in grasslands. Global Ecol. Biogeogr. 11, 79–86.

Gilmanov, T.G., Johnson, D.A., Saliendra, N.Z., 2003. Growing season CO_2 fluxes in a sagebrush-steppe ecosystem in Idaho: Bowen ratio/energy balance measurements and modeling. Basic Appl. Ecol. 4, 167–183.

Hall, D.O., Ojima, D.S., Parton, W.J., Scurlock, J.M.O., 1995. Response of temperate and tropical grasslands to CO_2 and climate change. J. Biogeogr. 22, 537–547.

Hanson, P.J., Edwards, N.T., Garten, C.T., Andrews, J.A., 2000. Separating root and soil microbial contributions to soil respiration: a review of methods and observations. Biogeochemistry 48, 115–146.

Jawson, M.D., Shafer, S.R., Faranzluebbers, A.J., Parkin, T.B., Follett, R.F., 2005. GRACEnet: greenhouse gas reduction through agricultural carbon enhancement network. Soil Till. Res. 83, 167–172.

McCree, K.J., Troughton, J.H., 1966. Non-existence of an optimum leaf area index for the production rate of white clover grown under constant conditions. Plant Physiol. 41, 1615–1622.

Milchunas, D.G., Lauenroth, W.K., 1992. Carbon dynamics and estimates of primary production by harvest, ^{14}C dilution, and ^{14}C turnover. Ecology 73, 593–607.

Mohtar, R.H., Buckmaster, D.R., Fales, S.L., 1997. A grazing simulation model: GRASIM A: model development. Trans. ASAE 40, 1483–1493.

Nouvellon, Y., Moran, M.S., Lo Seen, D., Bryant, R., Rambal, S., Ni, W., Begue, A., Chehbouni, A., Emmerich, W.E., Heilman, P., Qi, J., 2001. Coupling a grassland ecosystem model with Landsat imagery for a 10-year simulation of carbon and water budgets. Remote Sens. Environ. 78, 131–149.

Passioura, J.B., 1996. Simulation models: science, snake oil, education, or engineering. Agron J. 88, 690–694.

Rotz, C.A., Corson, M.S., Coiner, C.U., 2007. The Integrated Farm System Model Reference Manual, Version 2.1. USDA-Agricultural Research Service. 141 p.

Saggar, S., Hedley, C., Mackay, A.D., 1997. Partitioning and translocation of photosynthetically fixed ^{14}C in grazed hill pastures. Biol. Fertil. Soils 25, 152–158.

Sanderson, M.A., Soder, K.J., Muller, L.D., Klement, K.D., Skinner, R.H., Goslee, S.C., 2005. Forage mixture productivity and botanical composition in pastures grazed by daily cattle. Agron. J. 97, 1465–1471.

Schlapfer, B., Ryser, P., 1996. Leaf and root turnover of three ecologically contrasting grass species in relation to their performance along a productivity gradient. Oikos 75, 398–406.
Skinner, R.H., Corson, M.S., Gilmanov, T.G., 2008. Simulating gross primary productivity of humid-temperate pastures. Agron. J. 100, 801–807.
Stewart, D.P.C., Metherell, A.K., 1999. Carbon (^{13}C) uptake and allocation in pasture plants following field pulse-labelling. Plant Soil 210, 61–73.
Thornley, J.H.M., Cannell, M.G.R., 1997. Temperate grassland responses to climate change: an analysis using the Hurley Pasture Model. Ann. Bot. 80, 205–221.
Thornley, J.H.M., Cannell, M.G.R., 2000. Modeling the components of plant respiration: representation and realism. Ann. Bot. 85, 55–67.
Van Noordwijk, M., Van de Geijn, S.C., 1996. Root, shoot and soil parameters required for process-oriented models of crop growth limited by water or nutrients. Plant Soil 183, 1–25.
Vico, G., Porporato, A., 2008. Modeling C_3 and C_4 photosynthesis under water-stressed conditions. Plant Soil. doi:10.1007/s11104-008-9691-4.
Weiss, A., Wilhelm, W.W., 2006. The circuitous path to the comparison of simulated values from crop models with field observations. J. Agric. Sci. 144, 475–488.
Willms, W.D., McGinn, S.M., Dormaar, J.F., 1993. Influence of litter on herbage production in the mixed prairie. J. Range Manage. 46, 320–324.